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Pitaval; Renaud-Alexandre et al.

DFT-s-OFDM SIGNAL WITH SPECTRUM EXTENSION

Abstract

Embodiments of the invention relate to DFT-s-OFDM signal with spectral extension. A communication device is provided that obtains N.sub.data Fourier coefficients based on N.sub.data data symbols, and repeats N.sub.e Fourier coefficients of the N.sub.data Fourier coefficients to obtain N.sub.sc Fourier coefficients, wherein N.sub.e is determined based on at least one of a modulation symbol constellation of the N.sub.data data symbols, and a frequency domain spectrum shaping (FDSS) window of the communication device. The N.sub.sc Fourier coefficients are multiplied with an FDSS window of size N.sub.sc to obtain N.sub.sc frequency shaped Fourier coefficients which are mapped onto N.sub.sc subcarriers to obtain a discrete Fourier transform spread orthogonal frequency division multiplexing (DFT-s-OFDM) signal, which is transmitted.

Inventors: Pitaval; Renaud-Alexandre (Kista, SE), Berggren; Fredrik (Kista, SE), Popovic; Branislav M. (Kista, SE)
Applicant: HUAWEI TECHNOLOGIES CO., LTD. (Shenzhen, CN)
Family ID: 83271376
Assignee: HUAWEI TECHNOLOGIES CO., LTD. (Shenzhen, CN)
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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of International Application No. PCT/EP2022/072944, filed on Aug. 17, 2022, the disclosure of which is hereby incorporated by reference in its entirety.

TECHNICAL FIELD

[0002] Embodiments of present disclosure relate to a first communication device and a second communication device for a discrete Fourier transform spread orthogonal frequency division multiplexing (DFT-s-OFDM) signal with spectral extension. Furthermore, embodiments of the present disclosure also relate to corresponding methods and a computer program.

BACKGROUND

[0003] Uplink transmissions are considered to be the coverage bottleneck of modern wireless systems due to limited user equipment (UE) transmission power. Therefore, uplink coverage enhancement is an important topic in current 3GPP new radio (NR) standardization. For coverage-limited scenario, the envelope fluctuation of the transmitted signals, typically measured by its cubic metric (CM) or peak-to-average power ratio (PAPR), should be as small as possible. This because a signal with higher CM/PAPR requires the power-amplifier to operate with larger power back-off in order to avoid signal distortion from its non-linear amplification range, directly impacting the link budget of the transmission.

[0004] Orthogonal frequency division multiplexing (OFDM) is to date the dominant modulating waveform of wireless communications systems, and the basis waveform of 3GPP standards in both the uplink and downlink. Discrete Fourier transform spread OFDM (DFT-s-OFDM) is a variant of OFDM principally used to achieve lower PAPR transmission than with standard OFDM. In DFT-s-OFDM the data symbols are first DFT-precoded before OFDM modulation, which leads to a form of single-carrier waveform as then the modulation symbols have their energy spread over all subcarrier in the allocated frequency spectrum. Reciprocally, in the time-domain, a large portion of a symbol energy is concentrated in short interval resulting in a time-domain multiplexing of symbols.

[0005] While PAPR is improved with DFT-s-OFDM, it can still be unsatisfactory, e.g., in deep-indoor coverage-limited scenario. Thus, additional low-PAPR techniques have been considered in 3GPP for uplink transmission. For example, 5G NR supports $\pi/2$ -BPSK in uplink in addition to legacy LTE constellations, aiming at providing further PAPR reduction and thus boosting radio frequency (RF) amplifier power efficiency at lower data-rates. In current 5G NR, $\pi/2$ -BPSK can also be used with frequency domain spectrum shaping (FDSS). FDSS enables also to bring additional PAPR

reduction at the cost of manageable self-interference. No specific FDSS window is defined in NR specification but is only indirectly permitted through looser RAN4 spectral flatness requirements specific for NR uplink with $\pi/2$ -BPSK.

[0006] Spectrum extension (SE) with FDSS is another PAPR reduction technique currently not specified in NR but listed as a potential solution. The main goal would be to achieve lower-PAPR also for high-order modulation such as QPSK and thus improving also the coverage for higher data rates. SUMMARY

[0007] The present disclosure provides a solution that mitigates or solves drawbacks and problems of conventional solutions.

[0008] The present disclosure provides a solution reducing PAPR compared to conventional solutions.

[0009] According to a first aspect, a first communication device for a communication system is provided, the first communication device being configured to [0010] obtain a N.sub.data number of Fourier coefficients based on a N.sub.data number of data symbols, wherein N.sub.data is a positive integer; [0011] repeat a N.sub.e number of Fourier coefficients of the N.sub.data number of Fourier coefficients to obtain a N.sub.sc number of Fourier coefficients, wherein N.sub.e and N.sub.sc are positive integers and wherein N.sub.e is determined based on at least one of: a modulation symbol constellation of the N.sub.data number of data symbols, and a frequency domain spectrum shaping, FDSS, window of the first communication device; [0012] multiply the N.sub.sc number of Fourier coefficients with a FDSS window of size N.sub.sc to obtain N.sub.sc number of frequency shaped Fourier coefficients; [0013] map the N.sub.sc number of frequency shaped Fourier coefficients on a N.sub.sc number of subcarriers to obtain a discrete Fourier transform spread orthogonal frequency division multiplexing, DFT-s-OFDM, signal; and transmit the DFT-s-OFDM signal.

[0014] An advantage of the first communication device according to the first aspect is that the first communication device allows the reduction of the PAPR of the DFT-s-OFDM signal. This is because the optimum reduction of PAPR and the value of N.sub.e to achieve it depends greatly on the modulation symbol constellation used for data symbols and the FDSS window. Thus, the first communication device as a transmitter device can use a lower power backoff, and thereby increase the coverage of the DFT-s-OFDM signal.

[0015] In an implementation form of a first communication device according to the first aspect, N.sub.e is determined further based on N.sub.sc or N.sub.data.

[0016] An advantage with this implementation form is further reduction of the PAPR of the DFT-s-OFDM signal according to the bandwidth allocation. This is because the value of N.sub.e to achieve the optimum reduction of PAPR roughly proportionally increase with the bandwidth allocation, but with slight variation.

[0017] In an implementation form of a first communication device according to the first aspect, the modulation symbol constellation of the N.sub.data number of data symbols is a $\pi/2$ -BPSK constellation or a QAM constellation.

[0018] An advantage with this implementation form is that such constellations are used in 3GPP standards of which $\pi/2$ -BPSK is specifically used to achieve a very low PAPR transmission which can be further improved by very different N.sub.e values than with QAM constellation.

[0019] In an implementation form of a first communication device according to the first aspect, [0020] N.sub.e is predetermined; or [0021] the first communication device is configured to receive a first control signal indicating N.sub.e.

[0022] An advantage with this implementation form is that having N.sub.e predetermined allows an easy implementation minimizing signaling overhead; or if N.sub.e is indicated by signaling it allows better reduction of PAPR according to other transmission parameters or device-specific characteristics.

[0023] In an implementation form of a first communication device according to the first aspect, the N.sub.e number of Fourier coefficient repetitions are: [0024] included in allocated resources for the transmission of the DFT-s-OFDM signal; or [0025] added to the allocated resources for the transmission of the DFT-s-OFDM signal.

[0026] An advantage with this implementation form is that with adjacent multiple user bandwidths, if including the repeated Fourier coefficients in the allocated resource of each user, avoids interference and the repeated symbols can then be used for improving demodulation performance; or if adding the repeated Fourier coefficients to the allocated resource the spectral efficiency is improved.

[0027] In an implementation form of a first communication device according to the first aspect, the first communication device being configured to transmit a second control signal indicating the capability of the first communication device of repeating the N.sub.e number of Fourier coefficients.

[0028] An advantage with this implementation form is that N.sub.e indicated by the first control signal can be determined based on the capability of the first communication device as indicated by second control signal. This allows optimized reduction of PAPR based on several configuration parameters and implementation aspects of the first communication device.

[0029] In an implementation form of a first communication device according to the first aspect, the first communication device being configured to, previous to multiplying the N.sub.sc number of Fourier coefficients with the FDSS window: [0030] cyclically shift the N.sub.sc number of Fourier coefficients with a L number of Fourier coefficients to obtain N.sub.sc number of cyclically shifted Fourier coefficients, wherein L is a positive integer; [0031] multiply the N.sub.sc number of cyclically shifted Fourier coefficients with the FDSS window of size N.sub.sc to obtain N.sub.sc number of frequency shaped and cyclically shifted Fourier coefficients; and [0032] map the N.sub.sc number of frequency shaped and cyclically shifted Fourier coefficients on the N.sub.sc number of subcarriers to obtain the DFT-s-OFDM signal.

[0033] An advantage with this implementation form is that it allows further reduction of PAPR as the PAPR changes as a function of the cyclically shift coefficient L.

[0034] In an implementation form of a first communication device according to the first aspect, [0035] L is predetermined; or [0036] the first communication device is configured to receive a first control signal indicating L.

[0037] An advantage with this implementation form is that having L predetermined allows an easy implementation minimizing signaling overhead; or if L is indicated by signaling it allows better reduction of PAPR according to other transmission parameters or device-specific characteristics.

[0038] In an implementation form of a first communication device according to the first aspect, L is determined based on any of: [0039] N.sub.e, N.sub.data or N.sub.sc; [0040] the modulation symbol constellation of the N.sub.data number of data symbols; and

[00001] $L = \text{round}(\frac{(2k+1)}{8}N_{\text{data}} - \frac{(N_e-1)}{2})$, or $L = \text{Math}.\frac{(2k+1)}{8}N_{\text{data}} - \frac{(N_e-1)}{2}.\text{Math}.$, or $L = \text{Math}.\frac{(2k+1)}{8}N_{\text{data}} - \frac{(N_e-1)}{2}.\text{Math}.$ or $L = \text{round}(N_{\text{data}} - \frac{(N_e-1)}{2})$, or $L = \text{Math}.\frac{(2k+1)}{8}N_{\text{data}} - \frac{(N_e-1)}{2}.\text{Math}.$, or $L = \text{Math}.\frac{(2k+1)}{8}N_{\text{data}} - \frac{(N_e-1)}{2}.\text{Math}.$, where k is a positive integer k, where round(x) gives the closest integer to x, and where $\lceil x \rceil$ and $\lfloor x \rfloor$ are the ceiling and floor operators on x, respectively.

[0041] An advantage with this implementation form is that it allows the best reduction of PAPR by selecting the best shifting function.

[0042] In an implementation form of a first communication device according to the first aspect, N.sub.e ≤ N.sub.e.sup.(opt) is less than a maximum allowed Fourier coefficient repetition capability N.sub.e.sup.(opt) of the first communication device (**100**), and wherein N.sub.e.sup.(opt) is determined based on any of: the FDSS window, L, and N.sub.sc and N.sub.data.

[0043] An advantage with this implementation form is that it allows selecting a smaller N.sub.e for improving spectral efficiency, and further avoids selecting a too large N.sub.e that could increase PAPR.

[0044] In an implementation form of a first communication device according to the first aspect, the first communication device being configured to determine N.sub.e ≤ N.sub.e.sup.(opt) to minimize a peak-to-average-power ratio, PAPR, of the transmission of the DFT-s-OFDM signal based on any of: the FDSS window, the modulation symbol constellation of the N.sub.data number of data symbols, L, and N.sub.sc or N.sub.data.

[0045] An advantage with this implementation form is that it allows selecting N.sub.e that optimize the PAPR performance according to transmission parameters or device-specific characteristics.

[0046] In an implementation form of a first communication device according to the first aspect, the first communication device being configured to

transmit a second control signal indicating N.sub.e.sup.(opt).

[0047] An advantage with this implementation form is that ensure that the selected N.sub.e is not detrimental in term of PAPR according to the specific first communication device capability and implementation.

[0048] According to a second aspect, a second communication device for a communication system is provided, the second communication device being configured to: [0049] receive a DFT-s-OFDM signal from a first communication device, the DFT-s-OFDM signal comprising a N.sub.sc number of Fourier coefficients mapped on a N.sub.sc number of subcarriers, wherein N.sub.sc is a positive integer; [0050] obtain the N.sub.sc number of Fourier coefficients based on the DFT-s-OFDM signal, the N.sub.sc number of Fourier coefficients comprising a N.sub.data number of Fourier coefficients and a N.sub.e number of repeated Fourier coefficients of the N.sub.data number of Fourier coefficients, wherein N.sub.data and N.sub.e are positive integers, and the N.sub.data number of Fourier coefficients are obtained based on a N.sub.data number of data symbols, and wherein N.sub.e is determined based on at least one of a modulation symbol constellation of the N.sub.data number of data symbols, and a FDSS window of the first communication device; [0051] obtain the N.sub.data number of Fourier coefficients based on the N.sub.sc number of Fourier coefficients; and [0052] decode the N.sub.data number of Fourier coefficients to obtain the N.sub.data number of data symbols.

[0053] An advantage of the second communication device according to the second aspect is that the first communication device allows the reduction of the PAPR of the DFT-s-OFDM signal. This is because the optimum reduction of PAPR and the value of N.sub.e to achieve it depends greatly on the modulation symbol constellation used for data symbols and the FDSS window. Thus, the first communication device as a transmitter device can use a lower power backoff, and thereby increase the coverage of the DFT-s-OFDM signal.

[0054] In an implementation form of a second communication device according to the second aspect, N.sub.e is determined further based on N.sub.sc or N.sub.data.

[0055] An advantage with this implementation form is further reduction of the PAPR of the DFT-s-OFDM signal according to the bandwidth allocation. This is because the value of N.sub.e to achieve the optimum reduction of PAPR roughly proportionally increase with the bandwidth allocation, but with slight variation.

[0056] In an implementation form of a second communication device according to the second aspect, the modulation symbol constellation of the N.sub.data number of data symbols is a $\pi/2$ -BPSK constellation or a QAM constellation.

[0057] An advantage with this implementation form is that such constellations are used in 3GPP standards of which $\pi/2$ -BPSK is specifically used to achieve a very low PAPR transmission which can be further improved by very different N.sub.e values that with QAM constellation.

[0058] In an implementation form of a second communication device according to the second aspect, [0059] N.sub.e is predetermined; or [0060] the second communication device is configured to transmit a first control signal indicating N.sub.e.

[0061] An advantage with this implementation form is that having N.sub.e predetermined allows an easy implementation minimizing signaling overhead; or if N.sub.e is indicated by signaling it allows better reduction of PAPR according to other transmission parameters or device-specific characteristics.

[0062] In an implementation form of a second communication device according to the second aspect, the N.sub.e number of Fourier coefficient repetitions are: [0063] included in allocated resources for the transmission of the DFT-s-OFDM signal; or [0064] added to the allocated resources for the transmission of the DFT-s-OFDM signal.

[0065] An advantage with this implementation form is that with adjacent multiple user bandwidths, if including the repeated Fourier coefficients in the allocated resource of each user, avoids interference and the repeated symbols can then be used for improving demodulation performance; or if adding the repeated Fourier coefficients to the allocated resource the spectral efficiency is improved.

[0066] In an implementation form of a second communication device according to the second aspect, [0067] the second communication device being configured to [0068] receive a second control signal indicating the capability of the first communication device of repeating the N.sub.e number of Fourier coefficients.

[0069] An advantage with this implementation form is that N.sub.e indicated by the first control signal can be determined based on the capability of the first communication device as indicated by second control signal. This allows optimized reduction of PAPR based on several configuration parameters and implementation aspects of the first communication device.

[0070] In an implementation form of a second communication device according to the second aspect, the second communication device being configured to, previous to obtain the N.sub.data number of Fourier coefficients: [0071] cyclically shift the N.sub.sc number of Fourier coefficients with a L number of Fourier coefficients to obtain N.sub.sc number of cyclically shifted Fourier coefficients, wherein L is a positive integer; and [0072] obtain the N.sub.data number of Fourier coefficients based on the N.sub.sc number of cyclically shifted Fourier coefficients.

[0073] An advantage with this implementation form is that it allows further reduction of PAPR as the PAPR changes as a function of the cyclically shift coefficient L.

[0074] In an implementation form of a second communication device according to the second aspect, [0075] L is predetermined; or [0076] the second communication device is configured to transmit a second control signal indicating L.

[0077] An advantage with this implementation form is that having L predetermined allows an easy implementation minimizing signaling overhead; or if L is indicated by signaling it allows better reduction of PAPR according to other transmission parameters or device-specific characteristics.

[0078] In an implementation form of a second communication device according to the second aspect, L is determined based on any of: [0079] N.sub.e, N.sub.data or N.sub.sc; [0080] the modulation symbol constellation of the N.sub.data number of data symbols; and [0081] formulas [00002]

$$L = \text{round}(\frac{(2k+1)}{8}N_{\text{data}} - \frac{(N_e-1)}{2}), \text{ or } L = \lceil \text{Math.}(\frac{(2k+1)}{8}N_{\text{data}} - \frac{(N_e-1)}{2}) \rceil, \text{ or } L = \lfloor \text{Math.}(\frac{(2k+1)}{8}N_{\text{data}} - \frac{(N_e-1)}{2}) \rfloor, \text{ or } L = \text{round}(N_{\text{data}} - \frac{(N_e-1)}{2}), \text{ or } L = \lceil N_{\text{data}} - \frac{(N_e-1)}{2} \rceil, \text{ or } L = \lfloor N_{\text{data}} - \frac{(N_e-1)}{2} \rfloor,$$
 where k is a positive integer k, where round(x) gives the closest integer to x, and where $\lceil x \rceil$ and $\lfloor x \rfloor$ are the ceiling and floor operators on x, respectively.

[0082] An advantage with this implementation form is that it allows the best reduction of PAPR by selecting the best shifting function.

[0083] In an implementation form of a second communication device according to the second aspect, [0084] N.sub.e ≤ N.sub.e.sup.(opt) is less than a maximum allowed Fourier coefficient repetition capability N_{opt} of the first communication device (100), and wherein N.sub.e.sup.(opt) is determined based on any of: the FDSS window, L, and N.sub.sc and N.sub.data.

[0085] An advantage with this implementation form is that it allows selecting a smaller N.sub.e for improving spectral efficiency, and further avoids selecting a too large N.sub.e that could increase PAPR.

[0086] In an implementation form of a second communication device according to the second aspect, N.sub.e ≤ N.sub.e.sup.(opt) to minimize a PAPR of the transmission of the DFT-s-OFDM signal based on any of: the FDSS window, the modulation symbol constellation of the N.sub.data number of data symbols, L, and N.sub.sc or N.sub.data.

[0087] An advantage with this implementation form is that it allows selecting N.sub.e that optimize the PAPR performance according to transmission parameters or device-specific characteristics.

[0088] In an implementation form of a second communication device according to the second aspect, the second communication device being configured to [0089] receive a second control signal indicating N.sub.e.sup.(opt).

[0090] An advantage with this implementation form is that ensure that the selected N.sub.e is not detrimental in term of PAPR according to the specific first communication device capability and implementation.

[0091] According to a third aspect, a method for a first communication device is provided, the method comprising: [0092] obtaining a N.sub.data number of Fourier coefficients based on a N.sub.data number of data symbols, wherein N.sub.data is a positive integer; [0093] repeating a N.sub.e

number of Fourier coefficients of the N.sub.data number of Fourier coefficients, wherein N.sub.e and N.sub.sc are positive integers and wherein N.sub.e is determined based on at least one of: a modulation symbol constellation of the N.sub.data number of data symbols, and a FDSS window of the first communication device; [0094] multiplying the N.sub.sc number of Fourier coefficients with a FDSS window of size N.sub.sc to obtain N.sub.sc number of frequency shaped Fourier coefficients; [0095] mapping the N.sub.sc number of frequency shaped Fourier coefficients on a N.sub.sc number of subcarriers to obtain a DFT-s-OFDM signal; and [0096] transmitting the DFT-s-OFDM signal.

[0097] The method according to the third aspect can be extended into implementation forms corresponding to the implementation forms of the first communication device according to the first aspect. Hence, an implementation form of the method comprises the feature(s) of the corresponding implementation form of the first communication device.

[0098] The advantages of the methods according to the third aspect are the same as those for the corresponding implementation forms of the first communication device according to the first aspect.

[0099] According to a fourth aspect, a method for a second communication device is provided, the method comprising: [0100] receiving a DFT-s-OFDM signal from a first communication device, the DFT-s-OFDM signal comprising a N.sub.sc number of Fourier coefficients mapped on a N.sub.sc number of subcarriers, wherein N.sub.sc is a positive integer; [0101] obtaining the N.sub.sc number of Fourier coefficients based on the DFT-s-OFDM signal, the N.sub.sc number of Fourier coefficients comprising a N.sub.data number of Fourier coefficients and a N.sub.e number of repeated Fourier coefficients of the N.sub.data number of Fourier coefficients, wherein N.sub.data and N.sub.e are positive integers, and the N.sub.data number of Fourier coefficients are obtained based on a N.sub.data number of data symbols, and wherein N.sub.e is determined based on at least one of a modulation symbol constellation of the N.sub.data number of data symbols, and a FDSS window of the first communication device; [0102] obtaining the N.sub.data number of Fourier coefficients based on the N.sub.sc number of Fourier coefficients; and [0103] decoding the N.sub.data number of Fourier coefficients to obtain the N.sub.data number of data symbols.

[0104] The method according to the fourth aspect can be extended into implementation forms corresponding to the implementation forms of the second communication device according to the second aspect. Hence, an implementation form of the method comprises the feature(s) of the corresponding implementation form of the second communication device.

[0105] The advantages of the methods according to the fourth aspect are the same as those for the corresponding implementation forms of the second communication device according to the second aspect.

[0106] Embodiments of the invention also relate to a computer program, characterized in program code, which when run by at least one processor causes the at least one processor to execute any method according to embodiments of the invention. Further, embodiments of the invention also relate to a computer program product comprising a computer readable medium and the mentioned computer program, wherein the computer program is included in the computer readable medium, and may comprises one or more from the group of: read-only memory (ROM), programmable ROM (PROM), erasable PROM (EPROM), flash memory, electrically erasable PROM (EEPROM), hard disk drive, etc.

[0107] Further applications and advantages of embodiments of the invention will be apparent from the following detailed description.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0108] The appended drawings are intended to clarify and explain different embodiments of the invention, in which:

[0109] FIG. 1 shows a first communication device according to embodiments of the invention;

[0110] FIG. 2 shows a flow chart of a method for a first communication device according to embodiments of the invention;

[0111] FIG. 3 shows a second communication device according to embodiments of the invention;

[0112] FIG. 4 shows a flow chart of a method for a second communication device according to embodiments of the invention;

[0113] FIG. 5 shows a communication system according to embodiments of the invention;

[0114] FIG. 6 shows a first implementation example of the first communication device with FDSS and spectrum extension with shift parameter L;

[0115] FIG. 7 shows an implementation example of the second communication device;

[0116] FIG. 8 (a) and FIG. 8 (b) and FIG. 8 (c) are a show 99-percentile PAPR [dB] as a function of N.sub.e for different constellations, FDSS window and values of L;

[0117] FIG. 9 shows 99-percentile PAPR [dB] of QPSK as a function of N.sub.e for a fixed N.sub.data=96;

[0118] FIG. 10(a) and FIG. 10(b) show optimal numbers N.sub.e.sup.(opt) of REs used for spectrum extension that minimizes the PAPR as a function of the number resource blocks N.sub.RB=N.sub.sc/12 for the total allocated bandwidth;

[0119] FIG. 11 shows optimal numbers N.sub.e.sup.(opt) of REs used for spectrum extension that minimizes the PAPR with QPSK where N.sub.data is fixed compared to the case where N.sub.sc is fixed;

[0120] FIG. 12 shows some signaling aspects of embodiments of the invention;

[0121] FIG. 13 illustrates spectrum-extended data sequence as a function of the shift parameter L with N.sub.data=10 symbols and N.sub.e=4;

[0122] FIG. 14 shows a second implementation example of the first communication device with FDSS and spectrum extension with shift parameter L;

[0123] FIG. 15 shows a third implementation example of the first communication device with FDSS and periodic spectrum extension with shift parameter L;

[0124] FIG. 16 shows DMRS subcarrier location for DFT-s-OFDM;

[0125] FIG. 17 illustrates the RE allocation with three different DMRS design options;

[0126] FIG. 18 shows CCDF of PAPR with the three different options of DMRS design, where N.sub.data=72 and N.sub.sc=96 with RRC window ($\rho=0.5$, $\beta=-0.65$); and

[0127] FIG. 19 shows maximum PAPR of Option C as a function of SE size with the three different FDSS window.

DETAILED DESCRIPTION

[0128] Currently in 3GPP NR, FDSS is only implicitly supported for $\pi/2$ -BPSK uplink transmission through RAN4 requirements limiting the distortion from a FDSS window. Moreover, in current NR, the FDSS window is proprietary and may be different for each UE. Therefore, the SE cannot be predetermined for any user equipment (UE), otherwise the size of the SE for a specific UE may be too large and detrimental in term of PAPR performance. At the same time the resource allocation and signal configuration for each UE is typically scheduled by the base station (BS).

[0129] Thus, embodiments of the invention relate to a first communication device **100** and a second communication device **300** solving the issue of conventional solutions, which do not provide the minimum PAPR, since the size of the SE is constant or not optimized with respect to PAPR performance and UE-specific transmission configuration and capability. Moreover, the issue of conventional solutions where the shift of the Fourier coefficients is not selected in order to minimize the PAPR is also solved.

[0130] FIG. 1 shows a first communication device **100** according to an embodiment of the invention. In the embodiment shown in FIG. 1, the first communication device **100** comprises a processor **102**, a transceiver **104** and a memory **106**. The processor **102** is coupled to the transceiver **104** and the memory **106** by communication means **108** known in the art. The first communication device **100** may be configured for wireless and/or wired communications in a communication system. The wireless communication capability may be provided with an antenna or antenna array **110** coupled to the transceiver **104**, while the wired communication capability may be provided with a wired communication interface **112** e.g., coupled to the

transceiver **104**.

[0131] The processor **102** may be referred to as one or more general-purpose central processing units (CPUs), one or more digital signal processor (DSPs), one or more application-specific integrated circuit (ASICs), one or more field programmable gate array (FPGAs), one or more programmable logic device, one or more discrete gate, one or more transistor logic device, one or more discrete hardware component, or one or more chipsets. The memory **106** may be a read-only memory, a random access memory (RAM), or a non-volatile RAM (NVRAM). The transceiver **304** may be a transceiver circuit, a power controller, or an interface providing capability to communicate with other communication modules or communication devices, such as network nodes and network servers. The transceiver **104**, memory **106** and/or processor **102** may be implemented in separate chipsets or may be implemented in a common chipset. That the first communication device **100** is configured to perform certain actions can in this disclosure be understood to mean that the first communication device **100** comprises suitable means, such as e.g., the processor **102** and the transceiver **104**, configured to perform the actions.

[0132] According to embodiments of the invention, the first communication device **100** is configured to obtain a N.sub.data number of Fourier coefficients based on a N.sub.data number of data symbols, wherein N.sub.data is a positive integer. The first communication device **100** is further configured to repeat a N.sub.e number of Fourier coefficients of the N.sub.data number of Fourier coefficients to obtain a N.sub.sc number of Fourier coefficients, wherein N.sub.e and N.sub.sc are positive integers and wherein N.sub.e is determined based on at least one of: a modulation symbol constellation of the N.sub.data number of data symbols, and a frequency domain spectrum shaping, FDSS, window of the first communication device **100**. The first communication device **100** is further configured to multiply the N.sub.sc number of Fourier coefficients with a FDSS window of size N.sub.sc to obtain N.sub.sc number of frequency shaped Fourier coefficients. The first communication device **100** is further configured to map the N.sub.sc number of frequency shaped Fourier coefficients on a N.sub.sc number of subcarriers to obtain a discrete Fourier transform spread orthogonal frequency division multiplexing, DFT-s-OFDM, signal **510**. The first communication device **100** is further configured to transmit the DFT-s-OFDM signal **510**.

[0133] Furthermore, in an embodiment of the invention, the first communication device **100** for a communication system **500** comprises: a processor configured to: obtain a N.sub.data number of Fourier coefficients based on a N.sub.data number of data symbols, wherein N.sub.data is a positive integer; repeat a N.sub.e number of Fourier coefficients of the N.sub.data number of Fourier coefficients to obtain a N.sub.sc number of Fourier coefficients, wherein N.sub.e and N.sub.sc are positive integers and wherein N.sub.e is determined based on at least one of: a modulation symbol constellation of the N.sub.data number of data symbols, and a FDSS window of the first communication device **100**; multiply the N.sub.sc number of Fourier coefficients with a FDSS window of size N.sub.sc to obtain N.sub.sc number of frequency shaped Fourier coefficients; map the N.sub.sc number of frequency shaped Fourier coefficients on a N.sub.sc number of subcarriers to obtain a DFT-s-OFDM signal **510**; and a transceiver configured to: transmit the DFT-s-OFDM signal **510**.

[0134] Moreover, in yet another embodiment of the invention, the first communication device **100** for a communication system **500** comprises a processor and a memory having computer readable instructions stored thereon which, when executed by the processor, cause the processor to: obtain a N.sub.data number of Fourier coefficients based on a N.sub.data number of data symbols, wherein N.sub.data is a positive integer; repeat a N.sub.e number of Fourier coefficients of the N.sub.data number of Fourier coefficients to obtain a N.sub.sc number of Fourier coefficients, wherein N.sub.e and N.sub.sc are positive integers and wherein N.sub.e is determined based on at least one of: a modulation symbol constellation of the N.sub.data number of data symbols, and a FDSS window of the first communication device **100**; multiply the N.sub.sc number of Fourier coefficients with a FDSS window of size N.sub.sc to obtain N.sub.sc number of frequency shaped Fourier coefficients; map the N.sub.sc number of frequency shaped Fourier coefficients on a N.sub.sc number of subcarriers to obtain a DFT-s-OFDM signal **510**; and transmit the DFT-s-OFDM signal **510**.

[0135] FIG. 2 shows a flow chart of a corresponding method **200** which may be executed in a first communication device **100**, such as the one shown in FIG. 1. The method **200** comprises obtaining **202** a N.sub.data number of Fourier coefficients based on a N.sub.data number of data symbols, wherein N.sub.data is a positive integer. The method **200** further comprises repeating **204** a N.sub.e number of Fourier coefficients of the N.sub.data number of Fourier coefficients to obtain a N.sub.sc number of Fourier coefficients, wherein N.sub.e and N.sub.sc are positive integers and wherein N.sub.e is determined based on at least one of: a modulation symbol constellation of the N.sub.data number of data symbols, and a FDSS window of the first communication device **100**. The method **200** further comprises multiplying **206** the N.sub.sc number of Fourier coefficients with a FDSS window of size N.sub.sc to obtain N.sub.sc number of frequency shaped Fourier coefficients. The method **200** further comprises mapping **208** the N.sub.sc number of frequency shaped Fourier coefficients on a N.sub.sc number of subcarriers to obtain a DFT-s-OFDM signal **510**. The method **200** further comprises transmitting **210** the DFT-s-OFDM signal **510**.

[0136] FIG. 3 shows a second communication device **300** according to an embodiment of the invention. In the embodiment shown in FIG. 3, the second communication device **300** comprises a processor **302**, a transceiver **304** and a memory **306**. The processor **302** is coupled to the transceiver **304** and the memory **306** by communication means **308** known in the art. The second communication device **300** further comprises an antenna or antenna array **310** coupled to the transceiver **304**, which means that the second communication device **300** is configured for wireless communications in a communication system.

[0137] The processor **302** may be referred to as one or more general-purpose CPUs, one or more DSPs, one or more ASICs, one or more FPGAs, one or more programmable logic device, one or more discrete gate, one or more transistor logic device, one or more discrete hardware component, one or more chipset. The memory **306** may be a read-only memory, a RAM, or a NVRAM. The transceiver **104** may be a transceiver circuit, a power controller, or an interface providing capability to communicate with other communication modules or communication devices. The transceiver **304**, the memory **306** and/or the processor **302** may be implemented in separate chipsets or may be implemented in a common chipset. That the second communication device **300** is configured to perform certain actions can in this disclosure be understood to mean that the second communication device **300** comprises suitable means, such as e.g., the processor **302** and the transceiver **304**, configured to perform the actions.

[0138] According to embodiments of the invention, the second communication device **300** is configured to receive a DFT-s-OFDM signal **510** from a first communication device **100**, the DFT-s-OFDM signal **510** comprising a N.sub.sc number of Fourier coefficients mapped on a N.sub.sc number of subcarriers, wherein N.sub.sc is a positive integer. The second communication device **300** is further configured to obtain the N.sub.sc number of Fourier coefficients based on the DFT-s-OFDM signal **510**, the N.sub.sc number of Fourier coefficients comprising a N.sub.data number of Fourier coefficients and a N.sub.e number of repeated Fourier coefficients of the N.sub.data number of Fourier coefficients, wherein N.sub.data and N.sub.e are positive integers, and the N.sub.data number of Fourier coefficients are obtained based on a N.sub.data number of data symbols, and wherein N.sub.e is determined based on at least one of a modulation symbol constellation of the N.sub.data number of data symbols, and a FDSS window of the first communication device **100**. The second communication device **300** is further configured to obtain the N.sub.data number of Fourier coefficients based on the N.sub.sc number of Fourier coefficients. The second communication device **300** is further configured to decode the N.sub.data number of Fourier coefficients to obtain the N.sub.data number of data symbols.

[0139] Furthermore, in an embodiment of the invention, the second communication device **300** for a communication system **500** comprises: a transceiver configured to: receive a DFT-s-OFDM signal **510** from a first communication device **100**, the DFT-s-OFDM signal **510** comprising a N.sub.sc number of Fourier coefficients mapped on a N.sub.sc number of subcarriers, wherein N.sub.sc is a positive integer; and a processor configured to: obtain the N.sub.sc number of Fourier coefficients based on the DFT-s-OFDM signal **510**, the N.sub.sc number of Fourier coefficients comprising a N.sub.data number of Fourier coefficients and a N.sub.e number of repeated Fourier coefficients of the N.sub.data number of Fourier coefficients, wherein N.sub.data and N.sub.e are positive integers, and the N.sub.data number of Fourier coefficients are obtained based on a N.sub.data number of data symbols, and wherein N.sub.e is determined based on at least one of a modulation symbol constellation of the N.sub.data number of data symbols, and a FDSS window of the first communication device **100**; obtain the N.sub.data number of Fourier coefficients based on

the N.sub.sc number of Fourier coefficients; and decode the N.sub.data number of Fourier coefficients to obtain the N.sub.data number of data symbols.

[0140] Moreover, in yet another embodiment of the invention, the second communication device **300** for a communication system **500** comprises a processor and a memory having computer readable instructions stored thereon which, when executed by the processor, cause the processor to: receive a DFT-s-OFDM signal **510** from a first communication device **100**, the DFT-s-OFDM signal **510** comprising a N.sub.sc number of Fourier coefficients mapped on a N.sub.sc number of subcarriers, wherein N.sub.sc is a positive integer; obtain the N.sub.sc number of Fourier coefficients based on the DFT-s-OFDM signal **510**, the N.sub.sc number of Fourier coefficients comprising a N.sub.data number of Fourier coefficients and a N.sub.e number of repeated Fourier coefficients of the N.sub.data number of Fourier coefficients, wherein N.sub.data and N.sub.e are positive integers, and the N.sub.data number of Fourier coefficients are obtained based on a N.sub.data number of data symbols, and wherein N.sub.e is determined based on at least one of a modulation symbol constellation of the N.sub.data number of data symbols, and a FDSS window of the first communication device **100**; obtain the N.sub.data number of Fourier coefficients based on the N.sub.sc number of Fourier coefficients; and decode the N.sub.data number of Fourier coefficients to obtain the N.sub.data number of data symbols.

[0141] FIG. 4 shows a flow chart of a corresponding method **400** which may be executed in a second communication device **300**, such as the one shown in FIG. 3. The method **400** comprises receiving **402** a DFT-s-OFDM signal **510** from a first communication device **100**, the DFT-s-OFDM signal **510** comprising a N.sub.sc number of Fourier coefficients mapped on a N.sub.sc number of subcarriers, wherein N.sub.sc is a positive integer. The method **400** further comprises obtaining **404** the N.sub.sc number of Fourier coefficients based on the DFT-s-OFDM signal **510**, the N.sub.sc number of Fourier coefficients comprising a N.sub.data number of Fourier coefficients and a N.sub.e number of repeated Fourier coefficients of the N.sub.data number of Fourier coefficients, wherein N.sub.data and N.sub.e are positive integers, and the N.sub.data number of Fourier coefficients are obtained based on a N.sub.data number of data symbols, and wherein N.sub.e is determined based on at least one of a modulation symbol constellation of the N.sub.data number of data symbols, and a FDSS window of the first communication device **100**. The method **400** further comprises obtaining **406** the N.sub.data number of Fourier coefficients based on the N.sub.sc number of Fourier coefficients. The method **400** further comprises decoding **408** the N.sub.data number of Fourier coefficients to obtain the N.sub.data number of data symbols.

[0142] FIG. 5 shows a communication system **500** according to embodiments of the invention. The communication system **500** in the disclosed example comprises a first communication device **100** and a second communication device **300** configured to communicate and operate in the communication system **500**. For simplicity, the shown communication system **500** only comprises one first communication device **100** and one second communication device **300**. However, the communication system **500** may comprise any number of first communication devices **100** and any number of second communication devices **300** without deviating from the scope of the invention.

[0143] The first communication device **100** is configured to generate and transmit a DFT-s-OFDM signal **510** according to the disclosed solution in the communication system **500**. The second communication device **300** is thus configured to receive the DFT-s-OFDM signal **510** transmitted by the first communication device **100**. The mentioned communication system **500** is any suitable communication system such as 3GPP 5G NR or 6G. The first communication device **100** may also be denoted a transmitter device or simply a transmitter and may be a client device such as a UE.

Correspondingly, the second communication device **300** may also be denoted receiver device or simply a receiver and may be network access node such as a BS. The network access node may be connected to a network (NW) such as a core network via a communication interface. However, embodiments of the invention are also applicable when the transmitter is a base station and the receiver is a UE, i.e., the revers case. Moreover, embodiments of the invention are also applicable to other network nodes, such as repeaters, relays, etc. Furthermore, embodiments of the invention are further applicable to direct communication between UEs such as over the sidelink (SL) interface.

[0144] To provide even deeper understanding of the disclosed solution signal definitions are hereby presented in the following section in which we consider a DFT-s-OFDM signal $s[n]$ with SE which carries N.sub.data constellation symbols, using a total of $N_{sc}=N_{sub.data}+N_{sub.e}$ OFDM subcarriers where N.sub.e is a number of subcarriers used for symbol repetition only for the purpose of PAPR reduction. We assume for simplicity of illustration and in relation with the symmetric nature of practical FDSS window that N.sub.e is an even integer. Nevertheless, the disclosed solution is straightforwardly generalizable to when N.sub.e is odd, by assuming e.g., a left SE of $[N_{sub.e}/2]$ and right SE of $[N_{sub.e}/2]$, or vice-versa.

[0145] The low-pass equivalent time-discrete DFT-s-OFDM signal representation is defined for samples $0 \leq n \leq N_{sub.fft}-1$ by

$$[00003] \quad s[n] = \frac{1}{\sqrt{N_{sc}N_{data}}} \sum_{k=0}^{N_{sc}-1} \text{Math. } W[k]X[k + L(\text{mod } N_{data})]e^{j\frac{2\pi}{N_{sc}}nk} \quad (1) \quad [0146] \text{ with frequency-domain symbols}$$

$$[00004] \quad X[k] = \frac{1}{\sqrt{N_{data}}} \sum_{m=0}^{N_{data}-1} \text{Math. } x[m]e^{j\frac{2\pi}{N_{data}}km} \quad (2) \quad [0147] \text{ where [0148] } N_{sub.fft} \text{ is the IFFT size of the OFDM modulation. [0149] } N_{sub.data} \text{ is the}$$

number of modulation constellation symbols. [0150] $N_{sub.sc}=N_{sub.e}+N_{sub.data}$ is the total number of OFDM subcarriers. [0151] N.sub.e is the number of subcarriers used for SE, which we refer also to as the size of SE. [0152] $\{W[0], \dots, W[N_{sub.sc}-1]\}$ are the FDSS window coefficients. [0153] L is the Fourier coefficient shift parameter. [0154] $(\text{mod } N)$ is the modulo-N operator.

[0155] It may be noted that other types of normalization factors than and

$$[00005] \quad \frac{1}{\sqrt{N_{sc}N_{data}}} \text{ and } \frac{1}{\sqrt{N_{data}}}$$

could be used in Eq. (1) and (2). Typically, $W[k]$ are real-valued, but embodiments of the invention are not limited to this and complex-valued FDSS could be used. Embodiments of the invention further includes the case where $W[k]=1$.

[0156] Thus, according to embodiments of the invention cyclically shifting with shifting parameter L may also introduced. This means that the first communication device **100** is according to embodiments configured to, previous to multiplying the N.sub.sc number of Fourier coefficients with the FDSS window: cyclically shift the N.sub.sc number of Fourier coefficients with a L number of Fourier coefficients to obtain N.sub.sc number of cyclically shifted Fourier coefficients, wherein L is a positive integer; and to multiply the N.sub.sc number of cyclically shifted Fourier coefficients with the FDSS window of size N.sub.sc to obtain N.sub.sc number of frequency shaped and cyclically shifted Fourier coefficients. The first communication device **100** finally maps the N.sub.sc number of frequency shaped and cyclically shifted Fourier coefficients on the N.sub.sc number of subcarriers to obtain the DFT-s-OFDM signal **510**.

[0157] Correspondingly, the second communication device **300** is according to embodiments configured to, previous to obtain the N.sub.data number of Fourier coefficients: cyclically shift (in the opposite direction to the direction done in the first communication device **100**) the N.sub.sc number of Fourier coefficients with a L number of Fourier coefficients to obtain N.sub.sc number of cyclically shifted Fourier coefficients, wherein L is a positive integer; and to obtain the N.sub.data number of Fourier coefficients based on the N.sub.sc number of cyclically shifted Fourier coefficients.

[0158] A first implementation example of such as first communication device **100** is shown in the block diagram of FIG. 6. Further implementation examples of the first communication device **100** are presented in the following sections. With reference to FIG. 6, the first communication device **100** comprises a N.sub.data-point DFT block **120** configured to receive a N.sub.data number of data symbols $x[m]$ and output N.sub.data number of Fourier coefficients based on the N.sub.data number of data symbols. The first communication device **100** further comprises a SE block **122** which is configured to repeat a N.sub.e number of Fourier coefficients of the N.sub.data number of Fourier coefficients to provide a N.sub.sc number of Fourier coefficients which are fed to the cyclically shifting block **124**. The N.sub.sc number of Fourier coefficients are thus cyclically shifted in the cyclically shifting block **124** with a L number of Fourier coefficients to output N.sub.sc number of cyclically shifted Fourier coefficients which are fed to the FDSS block **126**. The N.sub.sc number of cyclically shifted Fourier coefficients are thereafter multiplied with the FDSS window of size N.sub.sc in the FDSS block **126** to output N.sub.sc number of frequency shaped and cyclically shifted Fourier coefficients $X'[k]$ which are fed to the mapper block **128**. In the mapper block **128**, the N.sub.sc number of frequency shaped and cyclically shifted Fourier coefficients are mapped on a

N.sub.sc number of subcarriers among N.sub.fft subcarriers to output a DFT-s-OFDM signal of N.sub.fft subcarriers in the frequency domain which is converted into the time domain in the IFFT block **130**. Additionally, a cyclic prefix (CP) may be added in the CP block **132**. The DFT-s-OFDM signal with CP is thereafter transmitted in the communication system **500** as signal s[n].

[0159] An implementation example of a corresponding second communication device **300** is shown in the block diagram of FIG. 7. The received DFT-s-OFDM signal **510** is first demodulated by a CP-removal step in the CP removal block **320** and thereafter fed to the FFT block **322** which provides N.sub.fft subcarriers coefficients. The demapper block **324** thereafter selects N.sub.sc number of Fourier coefficients among these N.sub.fft subcarriers according to the bandwidth allocation of the first communication device **100**. The selected N.sub.sc Fourier coefficients are fed to the cyclical shifting block **326** to be cyclically shifted with a L number of Fourier coefficients to provide N.sub.data number of Fourier coefficients. A common method to mitigate the transmission channel fading and the FDSS window attenuation is to perform equalization on the N.sub.data number of Fourier coefficients which is performed in the equalization and combiner block **328**. For further improvement, and depending on whether the repeated symbols were included or added in the allocated resources for the transmission of the DFT-s-OFDM signal, the equalization and combiner block **328** may also be inputted with the N.sub.e number of Fourier coefficients corresponding to the repeated symbols. A Fourier coefficient received from the in-band and its repeated version received from the spectral extension can for example be co-phased and summed before equalization, thus providing combining and frequency-diversity gains. Finally, the N.sub.data number of equalized Fourier coefficients are fed to the N.sub.data-point inverse DFT (IDFT) block **330** which is configured to precode the N.sub.data number of equalized Fourier coefficients to provide N.sub.data number of demodulated data symbols which are sent to a decoder block for decoding (not shown in FIG. 7).

[0160] Given a FDSS window, both parameters N.sub.e and L will impact the PAPR performance. Here we disclose how to carefully choose these parameters to reach the best PAPR reduction. Specially, we will show that: [0161] 1. When using SE, i.e., N.sub.e>0, the PAPR typically decreases but only up to an optimum N.sub.e.sup.(opt) and then increases again. Therefore, selecting a too large N.sub.e can be detrimental not only for spectrum efficiency but also from a PAPR reduction perspective. This non-trivial optimum depends mainly on the FDSS window, but also on the shift value L, modulation symbol constellation of the N.sub.data number of data symbols, and bandwidth allocation. [0162] 2. The value of L providing the lowest PAPR depends on modulation symbol constellation of the N.sub.data number of data symbols, the extension size N.sub.e and the number of constellation symbols i.e., N.sub.data=N.sub.sc-N.sub.e.

Existence of a Global Optimal N.sub.e.sup.(Opt) Minimizing the PAPR

[0163] It is known that SE can further reduce the PAPR on top of FDSS. However, it is not known that there is an optimum SE size N.sub.e.sup.(opt) that minimizes the PAPR, and so that a too large SE may even increase the PAPR. Therefore, the SE size should be better selected such that N.sub.e≤N.sub.e.sup.(opt), i.e., less than N.sub.e.sup.(opt) which corresponds to a specific SE capability of the first communication device **100** as a transmitter. We will see that the exact N.sub.e.sup.(opt) specific to each transmitter depends on the FDSS window, L, and N.sub.sc and N.sub.data. Note that selecting N.sub.e≤N.sub.e.sup.(opt) may also be desired for mitigating the spectral efficiency reduction of SE at the cost of slightly larger PAPR. Selecting N.sub.e≤N.sub.e.sup.(opt) may also be necessary by the system configuration to e.g., constrain N.sub.e to be a factor of 12 subcarriers for having an integer number of RB.

[0164] In FIGS. **8 (a)-(c)**, we plot the 99-percentile PAPR as a function of the number of SE subcarriers N.sub.e while the total bandwidth is kept fixed to N.sub.sc=96. This is plotted for different filters, modulation symbol constellations and shift values L. In all these numerical comparison, two FDSS windows are considered i.e.,: i) a Kaiser window with β=3, and ii) a truncated RRC window with ρ=0.5, β=-0.65. The shift rules L=0 and

$$[00006] L = -\frac{N_e}{2}$$

from conventional solutions are used. We will show in the following sections why parameter

$$[00007] L = -\frac{N_e}{2}$$

is closest to the best shift method for π/2-BPSK. For QAM constellations, we use also

$$[00008] L = \left(\frac{N_{\text{data}}}{8} - \frac{N_e}{2} \right)$$

which we will also show later why it is close to the best shifting for QPSK.

[0165] FIGS. **8 (a)-(c)** show that the resulting PAPR is neither a linear nor a continuously decreasing function as a function of N.sub.e. There thus exists for all cases an optimum SE size N.sub.e.sup.(opt) that minimizes the PAPR from which a further increase of SE would become detrimental. Therefore, the SE size should preferably be selected such that N.sub.e≤N.sub.e.sup.(opt), where N.sub.e.sup.(opt) can be predetermined based on the transmission configuration such as the FDSS window, the shift parameter L, the modulation symbol constellation and the bandwidth allocation. It may also be noted from FIGS. **8 (a)-(c)** that even without FDSS, a large reduction of PAPR can be obtained from SE.

[0166] In simulations, for a given total bandwidth N.sub.sc we increased the value of N.sub.e and thus decreased the value of N.sub.data such that N.sub.e+N.sub.data=N.sub.sc. Therefore, the SE was performed inside the given total bandwidth of N.sub.sc subcarriers. Alternatively, one could consider optimizing the best N.sub.e such that we fix N.sub.data and increase the total bandwidth N.sub.sc. FIG. **9** compares the PAPR for different N.sub.e where N.sub.data=96 is fixed, with QPSK and

$$[00009] L = -\frac{N_e}{2}.$$

It can be observed from FIG. **9** that the curves look of similar shapes than for N.sub.sc fixed, and notably also show a global optimum.

FDSS Window and Convergence of N.sub.e.sup.(opt)/N.sub.sc as N.sub.sc Grows

[0167] The global optimum of SE size N.sub.e.sup.(opt) mainly depends on the FDSS window design. In fact, N.sub.e.sup.(opt) appears to converge to a percentage of N.sub.sc as N.sub.sc grows for a given FDSS window. This means that to achieve similar PAPR reduction N.sub.e should be linearly increased with the bandwidth allocation N.sub.sc or N.sub.data. Therefore N.sub.e must better be determined also a function of N.sub.sc, or equivalently N.sub.data. In other words, N.sub.e is in embodiments determined further based on parameters N.sub.sc or N.sub.data.

[0168] FIGS. **10 (a)** and **(b)** show the numerically found N.sub.e.sup.(opt) in percentage of N.sub.sc as a function of N.sub.RB=N.sub.sc/12 for different FDSS windows. The numerically found N.sub.e.sup.(opt) is expressed in percentage of N.sub.sc as a function of N.sub.RB=4, 8, 12, 24, 48, 96, and 100. For a given modulation symbol constellation, one can see that the optimal N.sub.e.sup.(opt) may be very different depending on the FDSS window. The lowest possible PAPR is obtained for values of N.sub.e that ranges in these examples between 7%-35% of the total bandwidth allocation. In general, the more edge attenuation a FDSS window provides, the lower the PAPR is and the largest optimum value N.sub.e.sup.(opt) is. More importantly, for each FDSS window, the optimum SE size N.sub.e.sup.(opt) appears to be almost a constant fraction of N.sub.sc as N.sub.sc grows.

[0169] Moreover, the obtained optimum N.sub.e.sup.(opt) with N.sub.data fixed is plotted in FIG. **11** as a percentage of the resulting N.sub.sc=N.sub.data+N.sub.e.sup.(opt). Here QPSK modulation is used with

$$[00010] L = -\frac{N_e}{2}$$

and a FDSS Kaiser window with β=2. This is compared to the obtained N.sub.e.sup.(opt) as a percentage of N.sub.sc when N.sub.sc is fixed. As it can be observed the optimization N.sub.e.sup.(opt) with N.sub.data fixed leads to very similar ratio N.sub.e.sup.(opt)/(N.sub.data+N.sub.e.sup.(opt)) compared to the optimization N.sub.e.sup.(opt)/N.sub.sc with N.sub.sc fixed. It is also noted that in the case of N.sub.data fixed, it could be more natural to express the optimum SE size as the ratio N.sub.e.sup.(opt)/N.sub.data with would be obviously a different ratio than N.sub.e.sup.(opt)/N.sub.sc. In general, the two ratios are always related as follows: assume

$$[00011] \frac{N_e^{(\text{opt})}}{N_{\text{sc}}} = X$$

then equivalently

$$[00012] \frac{N_e^{(\text{opt})}}{N_{\text{data}}} = \frac{X}{1-X}.$$

Semi-Analytical Approximation of N.sub.e.sup.(opt)

[0170] The optimum SE size N.sub.e.sup.(opt) minimizing the PAPR can also be verified and approximated by a semi-analytical numerical search. [0171] Thus, we start by reformulating the signal as a time-duplexing of multiple pulses {g.sub.m[n]}.sub.m=0.sup.N.sub.data.sup.-1. By inserting Eq. (2) in (1), we obtain,

$$[00013] \quad s[n] = \frac{1}{\sqrt{N_{sc}N_{data}}} \sum_{m=0}^{N_{data}-1} \text{Math. } x[m] h[n - \frac{N_{fft}}{N_{data}}m] e^{-j \frac{2\pi}{N_{data}}Lm} \quad (3) \quad [0172] \text{ where the pulse-shaping filter is:}$$

$$[00014] \quad h[n] = \sum_{k=0}^{N_{sc}-1} \text{Math. } W[k] e^{j \frac{2\pi}{N_{fft}}kn} \quad (4)$$

[0173] In the case of no FDSS windowing, W[0]=W[N.sub.sc-1]=1, this further reduces to the conventional DFT-s-OFDM pulse, in the form of a Dirichlet kernel, with N.sub.sc modulated subcarriers:

$$[00015] \quad h[n] = e^{j \frac{2\pi}{N_{fft}}n(N_{sc}-1) \frac{\sin(\frac{N_{sc}n}{N_{fft}})}{\sin(\frac{n}{N_{fft}})}} \quad (5)$$

[0174] Using this pulse formulation and adapting the bound, we get the following bounds for ϕ -rotate BPSK on our system as

$$[00016] \quad \text{PAPR} \leq \max_n \left\{ \frac{N_{fft}N_{sc}}{N_{data}P_W} \sum_{i,j=0}^{N_{data}-1} \text{Math. } g_i[n] \text{Math. } g_j[n] \text{Math. } \cos\left(\frac{(i-j)\pi}{N_{data}}\right) \right\} = \text{PAPR}^{\text{-BPSK}} \quad (6)$$

where P.sub.W= $\sum_{k=0}^{N_{sc}-1} |W[k]|^2$ is the norm square of the FDSS window.

[0175] For $\pi/2$ -BPSK, we specifically have $\phi=\pi/2$. Moreover, choosing $L=-N_{sub.e}/2$, which is the best shifting method, it follows that

$$[00017] \quad \frac{2L+N_e-1}{N_{data}} = \frac{-1}{N_{data}} \approx 0.$$

Hence, the bound of Eq. (6) become approximatively equal to

$$[00018] \quad \text{PAPR}^{\text{-BPSK}} \approx \max_n \left\{ \frac{N_{fft}N_{sc}}{N_{data}P_W} \sum_{i,j=0}^{N_{data}-1} \text{Math. } g_i[n] \text{Math. } g_j[n] \text{Math. } \cos\left(\frac{(i-j)\pi}{2}\right) \right\} = \max_n \left\{ \frac{N_{fft}N_{sc}}{N_{data}P_W} \sum_{\substack{i,j=0 \\ (i-j) \text{ odd}}}^{N_{data}-1} \text{Math. } g_i[n] \text{Math. } g_j[n] \right\}$$

[0176] Note that in the case N.sub.e is allowed to be an odd integer, equality would hold with $L=-(N_{sub.e}-1)/2$.

[0177] For QPSK, and in fact any PSK modulation, we can derive a similar bound. The bound becomes

$$[00019] \quad \text{PAPR} \leq \max_n \left\{ \frac{N_{fft}N_{sc}}{N_{data}P_W} \sum_{i,j=0}^{N_{data}-1} \text{Math. } g_i[n] \text{Math. } g_j[n] \right\} = \text{PAPR}^{\text{QPSK}} \quad (8)$$

[0178] By using $|\cos\theta| \leq 1$, on can directly verify that $\text{PAPR}^{\text{sup.}\phi\text{-BPSK}} \leq \text{PAPR}^{\text{sup.QPSK}}$. Moreover, if $\phi=0$ as in normal BPSK and if one can have $L=-(N_{sub.e}-1)/2$, these two bounds become the same: $\text{PAPR}^{\text{sup.}(\phi=0)\text{-BPSK}} = \text{PAPR}^{\text{sup.QPSK}}$.

[0179] In order to compute these bounds, we need to find a maximum over N.sub.fft time samples. Nevertheless, the computation complexity of this step can be greatly reduced by remarking that the functions inside the max-operator are periodic so that it is enough to compute only

$$[00020] \quad \text{Math. } \frac{N_{fft}}{N_{data}} \text{Math.}$$

samples to find the maximum value over n.

[0180] The bounds in Eq. (7) and (8) may be compared by simulated 99-percentile PAPR. The bounds in Eq. (7) and (8) are not tight, but nevertheless these bounds appear to preserve roughly the position of the optimum SE size minimizing the PAPR. Therefore, we can numerically search the value N.sub.e minimizing these bounds to approximate the optimum SE size. This is shown in FIGS. 10 (a) and (b) where we see that this semi-analytical approximation of N.sub.e.sup.(opt) is often very close and at worst within a 5% gap, except for the case of $\pi/2$ -BPSK with Kaiser window ($\beta=3$). For this specific case, the semi analytical search is very far away from the numerical search and appears to correspond to the location of the minimum PAPR with $L=0$, see FIG. 8(a). The issue in this case it that SE extension is able to only provide a decrease of PAPR of maximum 0.32 dB to the already very low PAPR of 1.84 dB without SE. The resulting bound is accordingly very flat for a large range of values without a very sharp minimum.

[0181] We provide also complementary intuitive explanation of the existence of this optimum SE size using the equivalent pulse-shaping formulation. The upper bound in Eq. (8) can be reformulated as

$$[00021] \quad \text{PAPR} \leq \frac{N_{fft}N_{sc}}{N_{data}P_W} \left(\max_n \left\{ \sum_{i=0}^{N_{data}-1} \text{Math. } g_i[n] \text{Math. } \right\} \right)^2 \quad (9)$$

[0182] The coefficient

$$[00022] \quad \frac{N_{fft}N_{sc}}{N_{data}P_W} = \frac{N_{fft}N_{sc}}{(N_{sc}-N_e)P_W} = \frac{N_{fft}(N_{data}+N_e)}{N_{data}P_W}$$

systematically increases when N.sub.e increases. However,

$$[00023] \quad \left(\max_n \left\{ \sum_{i=0}^{N_{data}-1} \text{Math. } g_i[n] \text{Math. } \right\} \right)^2$$

which is the square of the maximum coherent combining of the pulses, may decrease or increase as a function of N.sub.e. With a typical windowing function W[k] as described above, the pulse-shaping filters |g.sub.i[n]| remain essentially of sinc-shape but with more or less attenuated side lobes. The number of pulses is N.sub.data, as given from Eq. (3), and their time separation is

$$[00024] \quad \frac{N_{fft}}{N_{data}}$$

samples. Through Eq. (4), we further observe that the pulse shape only depends on the FDSS window and the total number of allocated subcarriers N.sub.sc. If we fix the value of N.sub.sc there will be N.sub.data=(N.sub.sc-N.sub.e) pulses separated apart by

$$[00025] \quad \frac{N_{fft}}{N_{data}} = \frac{N_{fft}}{N_{sc}-N_e}$$

samples, i.e., the pulse shape does not change by increasing N.sub.e but the time-separation between the pulses increases. How much the pulse lobes overlap depends on the FDSS design, in any case an increased time-separation will play on this overlap, helping to reduce the peak power of the signal.

[0183] If we would instead fix the value N.sub.data, N.sub.sc will increase as N.sub.e increases and there will be N.sub.data pulses separated apart by

$$[00026] \quad \frac{N_{fft}}{N_{data}}$$

samples. Reciprocally in this case, the pulse shape of Eq. (5) without FDSS will become narrower, since the width of the main lobe is

$$[00027] \quad 2 \frac{N_{fft}}{N_{sc}}$$

samples. The narrower pulses will also help to reduce the peak power of the signal. In any case, the FDSS window will have to be designed for a bandwidth of N.sub.sc subcarriers.

Modulation Symbol Constellation

[0184] In embodiments of the invention, the modulation symbol constellation of the N.sub.data number of data symbols is a $\pi/2$ -BPSK constellation or a QAM constellation. In the case of the modulation symbol constellations QPSK, 16-QAM and 64-QAM, the optimum N.sub.e.sup.(opt) appears

to be very close to each other when employing the same FDSS window as shown in FIG. 10(b). The largest difference is 1.5% between 16-QAM and QPSK with Kaiser window ($\beta=3$). For several cases, the difference is unnoticeable. Therefore, a good rule of thumb is to select for 16-QAM and 64-QAM the same $N_{sub.e}$ as for QPSK. Table 1 below highlights the difference between the selection of $N_{sub.e,sup.(opt)}$ with $\pi/2$ -BPSK and QAMs constellations.

TABLE-US-00001 TABLE 1 Numerical selection of $N_{sub.e,sup.(opt)}$ according to FIG. 10 $N_{sub.e,sup.(opt)}/N_{sub.sc}$ $\pi/2$ - BPSK QAMs no FDSS
7% 10% Kaiser window ($\beta = 2$) 11% 21% RRC ($\rho = 0.5$, $\beta = -0.65$) 16% 25% Kaiser window ($\beta = 3$) 14% 30%

[0185] As it can be seen from Table 1, the optimum PAPR with $\pi/2$ -BPSK is obtained with much smaller $N_{sub.e}$ value than with QPSK for the same FDSS window. A good rule of thumb for $\pi/2$ -BPSK to get significant PAPR reduction is to select

$$[00028] N_e \leq \frac{1}{2} N_e^{(opt, QPSK)}$$

where $N_{sub.e,sup.(opt, QPSK)}$ is the optimum for QPSK, because

$$[00029] \frac{1}{2} N_e^{(opt, QPSK)} \leq N_e^{(opt, \pi/2 - BPSK)}.$$

Therefore, in summary the selection of $N_{sub.e}$ for all constellations can be derived from one single value of $N_{sub.e,sup.(opt)}$.

Shift Parameter L

[0186] In FIG. 8 (a) and FIG. 8 (b) and FIG. 8 (c), we see that the shifting method

$$[00030] L = \left(\frac{N_{data}}{8} - \frac{N_e}{2} \right)$$

derived in this disclosure provides the best PAPR, and

$$[00031] L = -\frac{N_e}{2}$$

from conventional solutions the worst PAPR. One may find that these two shifting methods reach their optimum for a similar value of SE size $N_{sub.e,sup.(opt)}$. With

$$[00032] L = \left(\frac{N_{data}}{8} - \frac{N_e}{2} \right),$$

the optimum SE size can be slightly reduced by 1-2% of $N_{sub.sc}$ compared to

$$[00033] L = -\frac{N_e}{2}.$$

[0187] We now explain why the best shifting method is

$$[00034] L = -\frac{N_e}{2}$$

for $\pi/2$ -BPSK; and

$$[00035] L = \frac{1}{8} N_{data} - \frac{N_e}{2}$$

for QPSK and other QAMs constellation. Starting from the multicarrier signal in Eq. (3), the phase difference between two neighboring pulses of indices m and (m+1) is

$$[00036] \angle \frac{g_{m+1}[n]}{g_m[n]} = -\frac{2}{N_{data}} L + \angle \frac{h[n - \frac{N_m}{N_{data}}(m+1)]}{h[n - \frac{N_m}{N_{data}}m]} \quad (10)$$

[0188] Then, if the FDSS window is real and symmetric we have by using

$$[00037] \angle \frac{h[n - \frac{N_m}{N_{data}}(m+1)]}{h[n - \frac{N_m}{N_{data}}m]} = -\frac{(N_{sc} - 1)}{N_{data}} + \{0 \text{ or } \pi\} \quad (11)$$

[0189] This can also be directly verified to hold true for the special case of no FDSS of Eq. (2).

[0190] Therefore, we get the pulsed phase difference

$$[00038] \angle \frac{g_{m+1}[n]}{g_m[n]} = -\frac{2}{N_{data}} \left(L + \frac{(N_{sc} - 1)}{2} \right) \pmod{\pi} \\ = -\frac{2}{N_{data}} \left(L + \frac{(N_e - 1)}{2} \right) \pmod{\pi} \quad (12)$$

[0191] The purpose of the rotated-constellation design as $\pi/2$ -BPSK is to ensure that BPSK symbol of neighboring pulse are transmitted with a phase different of almost $\pi/2 \pmod{\pi}$, such their maximum power combining is minimized.

[0192] For $\pi/2$ -BPSK, the best value of L is found such that Eq. (12) is the closest to zero, i.e.,

$$[00039] L \approx -\frac{(N_e - 1)}{2}.$$

If $N_{sub.e}$ is an odd integer, then

$$[00040] L = -\frac{(N_e - 1)}{2}$$

provides the best performance. If $N_{sub.e}$ is an even integer, then

$$[00041] L = -\frac{N_e}{2}$$

as in [7] or

$$[00042] L = \frac{(N_e - 2)}{2}$$

provides the best performance. Therefore, as it was shown in FIG. 8(a), selecting a shifting method close to

$$[00043] L \approx -\frac{(N_e - 1)}{2},$$

provide a much better PAPR performance than $L=0$.

[0193] For QPSK and other QAMs, choosing

$$[00044] L \approx -\frac{(N_e - 1)}{2}$$

is sub-optimal and better PAPR can almost always be found with other L. Notably, the lowest PAPR is obtained by creating a phase difference between neighboring pulses close to $\pi/4 \pmod{\pi/2}$. Namely, one should get

$$[00045] -\frac{2}{N_{data}} \left(L + \frac{(N_e - 1)}{2} \right) \approx \frac{\pi}{4} \pmod{\frac{\pi}{2}}. \quad (13)$$

[0194] Therefore, the best PAPR can always be achieved by selecting L approximately equal to

$$[00046] L \approx \frac{(2k+1)}{8} N_{data} - \frac{(N_e - 1)}{2}. \quad (14)$$

where k is an integer as then the shift by L leads to a phase difference among data symbols which is always close to $\pi/4$.

[0195] This is why in FIGS. 8(b) and (c), the PAPR was the worst with

$$[00047] L = -\frac{N_e}{2}$$

and the best

$$[00048] L = \frac{N_{data}}{8} - \frac{N_e}{2}.$$

The other value from conventional solutions $L=0$ provides a trade-off between these two previous L values as it reduces to the other two methods when i) either $N_{sub.e}=0$, then

$$[00049] L = -\frac{N_e}{2} = 0,$$

or ii) either when $N_{sub.e}=N_{sub.sc}/5$ which is equivalent to

$$[00050] N_e = \frac{N_{data}}{4},$$

and then

$$[00051] L = \frac{N_{data}}{8} - \frac{N_e}{2} = 0.$$

Therefore, $L=0$ provides close to the best performance only when $N_{sub.e} \approx 20\%$ of $N_{sub.sc}$.

[0196] The gain of the using

$$[00052] L = \frac{N_{\text{data}}}{8} - \frac{N_e}{2}$$

compared to conventional solutions can be up to 0.6 dB specially when the SE size $N_{\text{sub.e}}$ is less than the optimum $N_{\text{sub.e.sup.(opt)}}$ as it can be seen from in FIGS. 8(b) and (c). We recall that selecting $N_{\text{sub.e}} \leq N_{\text{sub.e.sup.(opt)}}$ may actually be desired for example for mitigating the spectral efficiency reduction of SE or constraining $N_{\text{sub.e}}$ to be a factor of 12 subcarriers for having an integer number of RB.

[0197] In summary, in order to achieve the best PAPR reduction L should be different for some modulation symbol constellation as shown above between $\pi/2$ -BPSK and QAMs. L should also be determined based on $N_{\text{sub.e}}$ and depending on the modulation symbol constellation also possibly based on $N_{\text{sub.data}}$ or $N_{\text{sub.sc}}$. Precisely, with QAM, the best PAPR values are achieved with

[00053]

$$L = \text{round}\left(\frac{(2k+1)}{8}N_{\text{data}} - \frac{(N_e-1)}{2}\right), \text{ or } L = \left\lceil \frac{(2k+1)}{8}N_{\text{data}} - \frac{(N_e-1)}{2} \right\rceil, \text{ or } L = \left\lfloor \frac{(2k+1)}{8}N_{\text{data}} - \frac{(N_e-1)}{2} \right\rfloor, \text{ or } L = \text{round}\left(N_{\text{data}} - \frac{(N_e-1)}{2}\right), \text{ or } L = \left\lceil N_{\text{data}} - \frac{(N_e-1)}{2} \right\rceil, \text{ or } L = \left\lfloor N_{\text{data}} - \frac{(N_e-1)}{2} \right\rfloor,$$

where k is a positive integer k, where round(x) gives the closest integer to x, and where $\lceil x \rceil$ and $\lfloor x \rfloor$ are the ceiling and floor operators on x, respectively.

Signaling Aspects

[0198] FIG. 12 illustrates signaling aspects according to embodiments of the invention, since based on the gains presented in the previous sections, the disclosed solution assumes that the number of subcarriers for the SE, $N_{\text{sub.e}}$, is based on the FDSS capability of the first communication device 100, and possibly other transmission configurations such as but not limited to: [0199] The total number of allocated subcarriers, $N_{\text{sub.sc}}$, or the number of subcarriers containing data symbols, $N_{\text{sub.data}}$. [0200] The Fourier coefficient shift parameter of L subcarriers. [0201] The modulation symbol constellation.

[0202] Both the transmitter (e.g., UE) and the receiver (e.g., base station) need to be aware of the selected values of parameters $N_{\text{sub.e}}$ and L. Such information could be provided by control signaling between the first communication device 100 (i.e., the transmitter) and the second communication device 300 (i.e., the receiver). The control signaling may e.g., be performed through higher protocol layers such as radio resource control (RRC) signaling or medium access control (MAC) signaling. This is beneficial if the parameters do not need to change in a dynamic fashion. Another option is to signal through the physical layer, e.g., via the control channels. The benefit of this is that the parameters can be changed instantly. Combinations of higher layer signaling and physical layer signaling could also be used, e.g., certain values of $N_{\text{sub.e}}$ and L are configured by higher layers and physical layer signaling selects among these values.

[0203] With reference to step I in FIG. 12, the second communication device 300 transmits a first control signal 520 to the first communication device 100. The first control signal 520 indicates parameter $N_{\text{sub.e}}$ and/or L. In step II in FIG. 12, the first communication device 100 is configured to receive the first control signal 520 indicating $N_{\text{sub.e}}$ and/or L from the second communication device 300.

[0204] However, parameters $N_{\text{sub.e}}$ and/or L, or parts of the mentioned parameters, could also be provided implicitly. For example, the parameters $N_{\text{sub.e}}$ and L could be predetermined and defined by standards for different modulation symbol constellations, for different number of allocated subcarriers, for different parameters of a frequency domain filter, etc. This avoids the use of control signaling and thus reduces the overhead in the communication system 500.

[0205] Moreover, when the first communication device 100 is a UE, the UE should signal to the BS whether it supports the use of SE. This can be done by so called UE capability signaling. In other words, the first communication device 100 may be configured to transmit a second control signal 530 indicating the capability of the first communication device 100 of repeating the $N_{\text{sub.e}}$ number of Fourier coefficients as shown in step III in FIG. 12. The second communication device 300 in step IV in FIG. 12 receives the second control signal 530 and thereby derives the information about the repeating capability of the first communication device 100.

[0206] Furthermore, if the FDSS window is not given by the communication standard, the first communication device 100 (e.g., a UE) could signal to the second communication device 300 (e.g., a BS) a value $N_{\text{sub.e.sup.(opt)}}$ which is the maximum value of $N_{\text{sub.e}}$ beyond which PAPR reduction is not achieved. In this way, the BS can avoid determining a larger value of $N_{\text{sub.e}}$ than is needed.

[0207] It may be noted that the signaling of the first control signal 520 and the second control signal 530 may be performed in the reverse order, i.e., transmitting the first control signal 520 after the second control signal 530, without deviating from the scope of the disclosed solution. The first control signal 520 and the second control signal 530 may also be transmitted concurrently.

[0208] Communication resources for the UE to transmit on the uplink can either be preconfigured, e.g., by semi-persistent scheduling or configured grant. Alternatively, the BS may transmit via a physical downlink control channel (PDCCH) an uplink grant to the UE containing information about the transmission, including the allocated resources blocks for the mentioned transmission. In embodiments of the invention, the $N_{\text{sub.e}}$ subcarriers may be included in the allocated resource blocks for the transmission, i.e., the $N_{\text{sub.e}}$ number of Fourier coefficient repetitions are included in allocated resources for the transmission of the DFT-s-OFDM signal 510. In other embodiments of the invention, the $N_{\text{sub.e}}$ subcarriers may be added to the allocated resource blocks for the transmission, i.e., the $N_{\text{sub.e}}$ number of Fourier coefficient repetitions are added to the allocated resources for the transmission of the DFT-s-OFDM signal 510.

Implementation Examples

[0209] The previously discussed and presented DFT-s-OFDM signal may typically be implemented by a cascade of DFT-precoding, a periodic SE with shift, a FDSS window, and an OFDM modulation as follows.

[0210] DFT-precoding: the constellation symbols $\{x[0], \dots, x[N_{\text{sub.data}}-1]\}$ are first DFT-precoded leading to the Fourier coefficients for $k=0, 1, \dots, N_{\text{sub.data}}-1$:

$$[00054] X[k] = \frac{1}{\sqrt{N_{\text{data}}}} \sum_{m=0}^{N_{\text{data}}-1} x[m] e^{-j \frac{2\pi}{N_{\text{data}}} km} \quad (15)$$

[0211] The transmitter chain was illustrated in FIG. 6 where the constellation symbols $\{x[0], \dots, x[N_{\text{sub.data}}-1]\}$ are first DFT-precoded in the DFT block 120 leading to the Fourier coefficients $\{X[0], \dots, X[N_{\text{sub.data}}-1]\}$.

[0212] Periodic SE with shift in SE block 122 and cyclical shift block 124: this frequency-domain sequence is allocated to an extended spectrum of $N_{\text{sub.sc}} = N_{\text{sub.data}} + N_{\text{sub.e}}$ subcarriers using periodic repetition as

$$[00055] X^{(se)}[k] = X[k + L(\text{mod } N_{\text{data}})] \quad (16)$$

for $k=0, 1, \dots, N_{\text{sub.sc}}-1$ and where L is an integer shift. Note that this definition of SE is directly valid also for $N_{\text{sub.e}}$ odd.

[0213] FDSS: before transmission by OFDM modulation, FDSS is applied to the Fourier coefficients in the FDSS block 126 as

$$[00056] X'[k] = W[k] X^{(se)}[k] \quad (17)$$

where $\{W[0], \dots, W[N_{\text{sub.sc}}-1]\}$ is the FDSS window.

[0214] OFDM modulation: finally, the (normalized) time-discrete lowpass equivalent signal is obtained by $N_{\text{sub.fft}}$ -point IFFT for $n=0, 1, \dots, N_{\text{sub.fft}}-1$ in IFFT block 130 as:

$$[00057] s[n] = \frac{1}{\sqrt{N_{\text{sc}} N_{\text{data}}}} \sum_{k=0}^{N_{\text{sc}}-1} X'[k] e^{j \frac{2\pi}{N_{\text{sc}}} nk} \quad (18)$$

Se with Shift

[0215] Moreover, the steps of SE with cyclical shift can be implemented via different equivalent embodiments. Different examples of the transmitted frequency-domain sequence as a function of the shift parameter L are shown in FIG. 13. For any value of L the original sequence of symbol $\{X[0], \dots$

, $X[N_{\text{sub.data}}-1]\}$ is always included in the in-band spectrum up to a cyclic shift by

$$[00058] L - \frac{N_e}{2}$$

symbols; and the left-side excess-band symbols are always the repetition of symbols of the right-side in-band edge; and similarly for the right-side excess band. Several embodiments are disclosed for implementing the SE according to Eq. (12).

[0216] In a first exemplary implementation, as shown in FIG. 6, the SE can be implemented by appending at the end of the original $N_{\text{sub.data}}$ -long sequence $X[k]$ its first $N_{\text{sub.e}}$ symbols and then cyclically-shifting the extended $N_{\text{sub.sc}}$ -long sequence by L symbols.

[0217] In a second exemplary implementation, as shown in FIG. 14, the cyclic SE can be implemented by first cyclically-shifting $N_{\text{sub.data}}$ -long sequence $X[k]$ by L symbols and then appending after the end of this shifted sequence its first $N_{\text{sub.e}}$ symbols. Thus, the difference between the exemplary implementation in FIGS. 6 and 14 is the order of cyclical shifting and SE operations. In FIG. 14, the cyclical shifting is performed previous to the SE operation.

[0218] In a third exemplary implementation, as shown in FIG. 15 the SE can be implemented by first cyclically shifting the original $N_{\text{sub.data}}$ -long sequence $X[k]$ by

$$[00059] (L - \frac{N_e}{2})$$

symbols as

$$[00060] X^{(L)}[k] = X[k + L - \frac{N_e}{2}(\text{mod } N_{\text{data}})], \text{ for } k = 0, \dots, N_{\text{data}} - 1 \quad (19)$$

and then appending the first and last

$$[00061] \frac{N_e}{2}$$

symbols respectively after and before this shifted sequence $X_{\text{sup}}(L)[k]$, i.e.,

$$[00062] x^{(se)} = \{x^{(L)}[N_{\text{data}} - \frac{N_e}{2}], \dots, x^{(L)}[N_{\text{data}} - 1], x^{(L)}[0], \dots, x^{(L)}[N_{\text{data}} - 1], x^{(L)}[0], \dots, x^{(L)}[\frac{N_e}{2} - 1]\} \quad (20)$$

[0219] Thus, the differences between the exemplary implementation in FIGS. 6 and 15 are the order of cyclic shifting and SE operations. In FIG. 15, the cyclical shifting is performed previous to the SE operation as in FIG. 13. However, the SE is also performed in another way in FIG. 15 as in FIGS. 6 and 14. Thus, the third exemplary implementation with

$$[00063] L = N_{\text{data}} - \frac{N_e}{2} = -\frac{N_e}{2}(\text{mod } N_{\text{data}})$$

corresponds to the symmetrical SE of conventional solution, cf. $L=-2$ in FIG. 13. Note that this selection of shift parameter L depends only on $N_{\text{sub.e}}$ and actually is independent of $N_{\text{sub.data}}$ as L is computed in $N_{\text{sub.data}}$ -modulo arithmetic. Moreover, we remark that the selection of L is purely for convenience as it does not bring any additional benefit.

Dmrs Transmission

[0220] In 5G NR, demodulation reference symbols (DMRS) are multiplexed together with data symbols, referred as physical uplink shared channel (PUSCH) in NR, where typically, only few OFDM symbols carries DMRS, for example 1 out of 14.

[0221] In the case that DFT-s-OFDM is used, DMRS are time-multiplexed with PUSCH. In an OFDM symbol carrying DMRS, the DMRS sequence is inserted without DFT-precoding on every other subcarrier (called resource element (RE) in NR) out of the $N_{\text{sub.sc}}$ subcarrier of the full transmission bandwidth, while other REs are blocked for data transmission. Two examples of possible configurations are shown in FIG. 16.

[0222] When data transmission is modulated $\pi/2$ -BPSK, one type of DMRS sequence is DFT precoded pseudo-noise $\pi/2$ -BPSK. Therefore, the properties of SE discussed for data transmission directly extends directly to such DMRS sequences.

[0223] For other constellations, another type of DMRS sequence is Zadoff-Chu (ZC) sequence. A DMRS sequence of length $M_{\text{sub.ZC}}$ have elements obtained by cyclic-extension

$$[00064] Z[k] = Z[k(\text{mod } N_{\text{ZC}})], k = 0, \dots, M_{\text{ZC}} - 1, \quad (25)$$

of a ZC sequence

$$[00065] Z[k] = e^{j \frac{u k(k+1)}{N_{\text{ZC}}}}, k = 0, \dots, N_{\text{ZC}} - 1, \quad (26)$$

with largest prime length satisfying $N_{\text{sub.ZC}} \leq M_{\text{sub.ZC}}$. The possible root indices are $u=1, \dots, N_{\text{sub.ZC}}-1$.

[0224] Note that the ZC sequence is a constant amplitude zero autocorrelation sequence (CAZAC) sequence and the IDFT of a ZC sequence is also a CAZAC sequence. Because of this property one can anticipate that DMRS with ZC sequence is a type of DFT-s-OFDM transmission and therefore the properties of SE discussed for data transmission also extends directly to such DMRS sequences.

[0225] Since the FDSS window for data transmission is typically not known at the receiver, the DMRS sequence needs to be shaped by the same FDSS window as the data symbols. Three options can be considered depending on the receiver capability.

[0226] Option A: the second communication device 300 as a receiver is only capable to decode the in-band spectrum without extension, it follows that the DMRS sequence needs to be designed according to the in-band, i.e., $M_{\text{sub.ZC}} = N_{\text{sub.data}}/2$

[0227] Option B: the second communication device 300 as a receiver is able to exploit the reception of the excess band symbols, and the DMRS sequence needs to be designed according to the whole allocation bandwidth $M_{\text{sub.ZC}} = N_{\text{sub.sc}}/2$.

[0228] Option C is to accommodate the two types of receivers with the same DMRS design. The sequence is first designed according to the in-band and then spectrally-extended to the whole allocation band. The DMRS sequence design becomes

$$[00066] Z[k] = Z[L + k(\text{mod } N_{\text{ZC}})], k = 0, \dots, N_{\text{sc}} / 2 - 1 \quad (23)$$

where $N_{\text{sub.ZC}}$ is the largest prime length satisfying $N_{\text{sub.ZC}} \leq N_{\text{sub.data}}/2$. The sequence without SE starts in the in-band, for this a symmetric extension is used which is equivalent to shift it by $L = \text{round}((N_{\text{sub.data}} - N_{\text{sub.sc}})/4)$.

[0229] These three options of DMRS designs are illustrated in FIG. 17. Their different PAPR with an RRC window, $N_{\text{sub.sc}}=96$ and $N_{\text{sub.data}}=72$ is shown in FIG. 18. The choice of $N_{\text{sub.data}}$ in FIG. 18 corresponds to have an SE of 25% of $N_{\text{sub.sc}}$ which is the optimum for PAPR reduction of QPSK data transmission for this FDSS window. As it can be observed Option C has the largest PAPR but this should be balanced by the fact it generates more sequences: 31 with Option B compared to 23 with Option A and C. Otherwise, the SE DMRS design Option C decreases the PAPR of Option A.

[0230] Finally, in FIG. 19 we compare the maximum PAPR of Option C with $N_{\text{sub.sc}}=96$, different SE size $N_{\text{sub.e}} = (N_{\text{sub.sc}} - N_{\text{sub.data}})$ and three different FDSS windows. Interestingly, the maximum PAPR of DMRS with SE behaves similarly than with data transmission. We can observe the PAPR first decreases to a minimum and then increase again. The SE size providing the lowest PAPR with DMRS is close the same SE size minimizing the PAPR with QPSK data transmission.

[0231] A network access node herein may also be denoted as a radio network access node, an access network access node, an access point (AP), or a base station (BS), e.g., a radio base station (RBS), which in some networks may be referred to as transmitter, “gNB”, “gNodeB”, “eNB”, “eNodeB”, “NodeB” or “B node”, depending on the standard, technology and terminology used. The radio network access node may be of different classes or types such as e.g., macro eNodeB, home eNodeB or pico base station, based on transmission power and thereby the cell size. The radio network access node may further be a station, which is any device that contains an IEEE 802.11-conformant media access control (MAC) and physical layer (PHY) interface to the wireless medium (WM). The radio network access node may be configured for communication in 3GPP related long term evolution (LTE), LTE-advanced, fifth generation (5G) wireless systems, such as new radio (NR) and their evolutions, as well as in IEEE related Wi-Fi, worldwide interoperability for microwave access (WiMAX) and their evolutions.

[0232] A client device herein may be denoted as a user device, a user equipment (UE), a mobile station, an internet of things (IoT) device, a sensor

device, a wireless terminal and/or a mobile terminal, and is enabled to communicate wirelessly in a wireless communication system, sometimes also referred to as a cellular radio system. The UEs may further be referred to as mobile telephones, cellular telephones, computer tablets or laptops with wireless capability. The UEs in this context may be, for example, portable, pocket-storable, hand-held, computer-comprised, or vehicle-mounted mobile devices, enabled to communicate voice and/or data, via a radio access network (RAN), with another communication entity, such as another receiver or a server. The UE may further be a station, which is any device that contains an IEEE 802.11-conformant MAC and PHY interface to the WM. The UE may be configured for communication in 3GPP related LTE, LTE-advanced, 5G wireless systems, such as NR, and their evolutions, as well as in IEEE related Wi-Fi, WiMAX and their evolutions

[0233] Furthermore, any method according to embodiments of the invention may be implemented in a computer program, having code means, which when run by processing means causes the processing means to execute the steps of the method. The computer program is included in a computer readable medium of a computer program product. The computer readable medium may comprise essentially any memory, such as previously mentioned a ROM, a PROM, an EPROM, a flash memory, an EEPROM, or a hard disk drive.

[0234] Moreover, it should be realized that the first communication device **100** and the second communication device **300** comprise the necessary communication capabilities in the form of e.g., functions, means, units, elements, etc., for performing or implementing embodiments of the invention. Examples of other such means, units, elements and functions are: processors, memory, buffers, control logic, encoders, decoders, rate matchers, de-rate matchers, mapping units, multipliers, decision units, selecting units, switches, interleavers, de-interleavers, modulators, demodulators, inputs, outputs, antennas, amplifiers, receiver units, transmitter units, DSPs, TCM encoder, TCM decoder, power supply units, power feeders, communication interfaces, communication protocols, etc. which are suitably arranged together for performing the solution.

[0235] Therefore, the processor(s) of the first communication device **100** and the second communication device **300** may comprise, e.g., one or more instances of a CPU, a processing unit, a processing circuit, a processor, an ASIC, a microprocessor, or other processing logic that may interpret and execute instructions. The expression “processor” may thus represent a processing circuitry comprising a plurality of processing circuits, such as e.g., any, some or all of the ones mentioned above. The processing circuitry may further perform data processing functions for inputting, outputting, and processing of data comprising data buffering and device control functions, such as call processing control, user interface control, or the like.

[0236] Finally, it should be understood that the invention is not limited to the embodiments described above, but also relates to and incorporates all embodiments within the scope of the appended independent claims.

Claims

1. A communication device for a communication system comprising: processing circuitry configured to: obtain N.sub.data Fourier coefficients based on N.sub.data data symbols, wherein N.sub.data is a positive integer; repeat N.sub.e Fourier coefficients of the N.sub.data Fourier coefficients to obtain N.sub.sc Fourier coefficients, wherein N.sub.e and N.sub.sc are positive integers and wherein N.sub.e is determined based on at least one of: a modulation symbol constellation of the N.sub.data data symbols, and a frequency domain spectrum shaping-(FDSS) window of the communication device; multiply the N.sub.sc Fourier coefficients with an FDSS window of size N.sub.sc to obtain N.sub.sc frequency shaped Fourier coefficients; map the N.sub.sc frequency shaped Fourier coefficients on N.sub.sc subcarriers to obtain a discrete Fourier transform spread orthogonal frequency division multiplexing (DFT-s-OFDM) signal; and transmit the DFT-s-OFDM signal.

2. The communication device according to claim 1, wherein N.sub.e is determined further based on N.sub.sc or N.sub.data.

3. The communication device according to claim 1, wherein the modulation symbol constellation of the N.sub.data data symbols is a $\pi/2$ -BPSK constellation or a QAM constellation.

4. The communication device according to claim 1, wherein the N.sub.e Fourier coefficient repetitions are: included in allocated resources for the transmission of the DFT-s-OFDM signal; or added to the allocated resources for the transmission of the DFT-s-OFDM signal.

5. The communication device according to claim 1, the processing circuitry being further configured to, prior to multiplying the N.sub.sc Fourier coefficients with the FDSS window: cyclically shift the N.sub.sc Fourier coefficients with L Fourier coefficients to obtain N.sub.sc cyclically shifted Fourier coefficients, wherein L is a positive integer; multiply the N.sub.sc cyclically shifted Fourier coefficients with the FDSS window of size N.sub.sc to obtain N.sub.sc frequency shaped and cyclically shifted Fourier coefficients; and map the N.sub.sc frequency shaped and cyclically shifted Fourier coefficients on the N.sub.sc subcarriers to obtain the DFT-s-OFDM signal.

6. The communication device according to claim 5, wherein L is determined based on any of: N.sub.e, N.sub.data or N.sub.sc; the modulation symbol constellation of the N.sub.data number of data symbols; and formulas

$$L = \text{round}\left(\frac{(2k+1)}{8}N_{\text{data}} - \frac{(N_e-1)}{2}\right), \text{ or } L = \left\lceil \frac{(2k+1)}{8}N_{\text{data}} - \frac{(N_e-1)}{2} \right\rceil, \text{ or } L = \left\lfloor \frac{(2k+1)}{8}N_{\text{data}} - \frac{(N_e-1)}{2} \right\rfloor.$$

$$\text{or } L = \left\lceil \frac{(2k+1)}{8}N_{\text{data}} - \frac{(N_e-1)}{2} \right\rceil, \text{ or } L = \left\lfloor \frac{(2k+1)}{8}N_{\text{data}} - \frac{(N_e-1)}{2} \right\rfloor.$$

$$\text{or } L = \text{round}\left(N_{\text{data}} - \frac{(N_e-1)}{2}\right), \text{ or } L = \left\lceil N_{\text{data}} - \frac{(N_e-1)}{2} \right\rceil, \text{ or } L = \left\lfloor N_{\text{data}} - \frac{(N_e-1)}{2} \right\rfloor,$$
where k is a positive integer, where round(x) gives the closest integer to x, and where $\lceil x \rceil$ and $\lfloor x \rfloor$ are the ceiling and floor operators on x, respectively.

7. The communication device according to claim 1, wherein N.sub.e ≤ N.sub.e.sup.(opt) is less than a maximum allowed Fourier coefficient repetition capability N.sub.e.sup.(opt) of the communication device, and wherein N.sub.e.sup.(opt) is determined based on any of: the FDSS window, L, and/or N.sub.sc and N.sub.data.

8. The communication device according to claim 7, the processing circuitry being further configured to: determine N.sub.e ≤ N.sub.e.sup.(opt) to minimize a peak-to-average-power ratio (PAPR) of the transmission of the DFT-s-OFDM signal based on one or more of: the FDSS window, the modulation symbol constellation of the N.sub.data data symbols, L, and/or N.sub.sc or N.sub.data.

9. A communication device for a communication system comprising: processing circuitry configured to: receive a discrete Fourier transform spread orthogonal frequency division multiplexing (DFT-s-OFDM) signal from a further communication device, the DFT-s-OFDM signal comprising N.sub.sc Fourier coefficients mapped on N.sub.sc subcarriers, wherein N.sub.sc is a positive integer; obtain the N.sub.sc Fourier coefficients based on the DFT-s-OFDM signal, the N.sub.sc Fourier coefficients comprising N.sub.data Fourier coefficients and N.sub.e repeated Fourier coefficients of the N.sub.data Fourier coefficients, wherein N.sub.data and N.sub.e are positive integers, and wherein the N.sub.data Fourier coefficients are obtained based on N.sub.data data symbols, and wherein N.sub.e is determined based on at least one of a modulation symbol constellation of the N.sub.data data symbols, and a frequency domain spectrum shaping (FDSS) window of a further communication device; obtain the N.sub.data Fourier coefficients based on the N.sub.sc Fourier coefficients; and decode the N.sub.data Fourier coefficients to obtain the N.sub.data data symbols.

10. The communication device according to claim 9, wherein N.sub.e is determined further based on N.sub.sc or N.sub.data.

11. The communication device according to claim 9, wherein the modulation symbol constellation of the N.sub.data data symbols is a $\pi/2$ -BPSK constellation or a QAM constellation.

12. The communication device according to claim 9, wherein the N.sub.e Fourier coefficient repetitions are: included in allocated resources for the transmission of the DFT-s-OFDM signal; or added to the allocated resources for the transmission of the DFT-s-OFDM signal.

13. The communication device according to claim 9, the processing circuitry being further configured to: cyclically shift the N.sub.sc Fourier coefficients with L Fourier coefficients to obtain N.sub.sc cyclically shifted Fourier coefficients, wherein L is a positive integer; and obtain the N.sub.data Fourier coefficients based on the N.sub.sc cyclically shifted Fourier coefficients.

14. The second communication device according to claim 13, wherein L is determined based on one or more of: N.sub.e, N.sub.data, and/or N.sub.sc; the modulation symbol constellation of the N.sub.data data symbols; and/or formulas

$$L = \text{round}\left(\frac{(2k+1)}{8}N_{\text{data}} - \frac{(N_e-1)}{2}\right), \text{ or } L = \left\lceil \frac{(2k+1)}{8}N_{\text{data}} - \frac{(N_e-1)}{2} \right\rceil, \text{ or } L = \left\lfloor \frac{(2k+1)}{8}N_{\text{data}} - \frac{(N_e-1)}{2} \right\rfloor,$$

$$\text{or } L = \left\lceil \frac{(2k+1)}{8}N_{\text{data}} - \frac{(N_e-1)}{2} \right\rceil, \text{ or } L = \left\lfloor \frac{(2k+1)}{8}N_{\text{data}} - \frac{(N_e-1)}{2} \right\rfloor,$$

$$\text{or } L = \text{round}\left(N_{\text{data}} - \frac{(N_e-1)}{2}\right), \text{ or } L = \left\lceil N_{\text{data}} - \frac{(N_e-1)}{2} \right\rceil, \text{ or } L = \left\lfloor N_{\text{data}} - \frac{(N_e-1)}{2} \right\rfloor,$$
where k is a positive integer, where round(x) gives the closest integer to x, and where $\lceil x \rceil$ and $\lfloor x \rfloor$ are the ceiling and floor operators on x, respectively.

where k is a positive integer k , where $\text{round}(x)$ gives the closest integer to x , and where $\lceil x \rceil$ and $\lfloor x \rfloor$ are the ceiling and floor operators on x , respectively.

15. The second communication device according to claim 9, wherein $N_{\text{sub.e}} \leq N_{\text{sub.e.sup.(opt)}}$, wherein $N_{\text{sub.e.sup.(opt)}}$ is a maximum allowed Fourier coefficient repetition capability of the first communication device, and wherein $N_{\text{sub.e.sup.(opt)}}$ is determined based on one or more of: the FDSS window, L , and/or $N_{\text{sub.sc}}$ and $N_{\text{sub.data}}$.

16. The second communication device according to claim 15, wherein $N_{\text{sub.e}} \leq N_{\text{sub.e.sup.(opt)}}$ to minimize a peak-to-average-power ratio (PAPR) of the transmission of the DFT-s-OFDM signal based on one or more of: the FDSS window, the modulation symbol constellation of the $N_{\text{sub.data}}$ number of data symbols, L , and/or $N_{\text{sub.sc}}$ or $N_{\text{sub.data}}$.

17. A method for a communication device, the method comprising: obtaining $N_{\text{sub.data}}$ Fourier coefficients based on $N_{\text{sub.data}}$ data symbols, wherein $N_{\text{sub.data}}$ is a positive integer; repeating $N_{\text{sub.e}}$ Fourier coefficients of the $N_{\text{sub.data}}$ Fourier coefficients to obtain $N_{\text{sub.sc}}$ Fourier coefficients, wherein $N_{\text{sub.e}}$ and $N_{\text{sub.sc}}$ are positive integers and wherein $N_{\text{sub.e}}$ is determined based on at least one of: a modulation symbol constellation of the $N_{\text{sub.data}}$ data symbols and/or a frequency domain spectrum shaping (FDSS) window of the first communication device; multiplying the $N_{\text{sub.sc}}$ Fourier coefficients with the FDSS window of size $N_{\text{sub.sc}}$ to obtain $N_{\text{sub.sc}}$ frequency shaped Fourier coefficients; mapping the $N_{\text{sub.sc}}$ frequency shaped Fourier coefficients on $N_{\text{sub.sc}}$ subcarriers to obtain a discrete Fourier transform spread orthogonal frequency division multiplexing (DFT-s-OFDM) signal; and transmitting the DFT-s-OFDM signal.

18. The method according to claim 16, wherein $N_{\text{sub.e}}$ is determined further based on $N_{\text{sub.sc}}$ or $N_{\text{sub.data}}$.

19. A method for a communication device, the method comprising: receiving a discrete Fourier transform spread orthogonal frequency division multiplexing (DFT-s-OFDM) signal from a first communication device, the DFT-s-OFDM signal comprising $N_{\text{sub.sc}}$ Fourier coefficients mapped on $N_{\text{sub.sc}}$ subcarriers, wherein $N_{\text{sub.sc}}$ is a positive integer; obtaining the $N_{\text{sub.sc}}$ Fourier coefficients based on the DFT-s-OFDM signal, the $N_{\text{sub.sc}}$ Fourier coefficients comprising $N_{\text{sub.data}}$ Fourier coefficients and $N_{\text{sub.e}}$ repeated Fourier coefficients of the $N_{\text{sub.data}}$ Fourier coefficients, wherein $N_{\text{sub.data}}$ and $N_{\text{sub.e}}$ are positive integers, and the $N_{\text{sub.data}}$ Fourier coefficients are obtained based on $N_{\text{sub.data}}$ data symbols, and wherein $N_{\text{sub.e}}$ is determined based on at least one of a modulation symbol constellation of the $N_{\text{sub.data}}$ data symbols, and a frequency domain spectrum shaping (FDSS) window of the first communication device; obtaining the $N_{\text{sub.data}}$ Fourier coefficients based on the $N_{\text{sub.sc}}$ Fourier coefficients; and decoding the $N_{\text{sub.data}}$ Fourier coefficients to obtain the $N_{\text{sub.data}}$ data symbols.

20. The method according to claim 19, wherein $N_{\text{sub.e}}$ is determined further based on $N_{\text{sub.sc}}$ or $N_{\text{sub.data}}$.
